

From Studio Set to Render: A Collaborative Framework for Hybrid Media Education

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• **Applied computing**; • **Education**; **Collaborative learning**; • **Computing methodologies**; • **Rendering**; **Visibility determination**;

Keywords

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1 Introduction: The End of the Silo.

The modern production landscape no longer distinguishes between the "captured" and the "created." From Virtual Production volumes to high-end commercial retouching, the tools are merging. However, a gap remains in education: 3D students often lack a physical "eye" for light, and lens-based students often lack the technical vocabulary to communicate with a VFX house. This talk details a course designed to bridge this divide by treating the classroom as a professional production studio.

2 Methodology: The Hybrid Production Workflow.

The course is structured around teams of three: a photographer/filmmaker, a 3D artist, and a compositor. Each project simulates a real-world brief—such as a large-scale automotive shoot or a complex product integration—that requires a seamless blend of live-action and CGI. The core of this methodology is the **Team Contract**, a technical service-level agreement that mirrors professional studio hand-offs.

- **Key Pedagogical Pillars:**
- **Shared Language:** We break down the barriers between physical camera terms (e.g., bokeh, depth of field, motion blur) and their digital equivalents in 3D engines.
- **Practical Realism:** 3D students are taken out of the computer lab and onto the photo set. By physically manipulating

lights and lenses, they gain an intuitive understanding of light decay and lens distortion that they can then "back-port" into their digital scenes.

- **The Collaborative Pipeline:** Students are forced to solve "hand-off" problems. For example, the lens-based students learn to capture HDRIs and provide lens data to the 3D team to ensure the CGI matches the physical environment perfectly.

3 Case Study: "SPACE METAL" Integration.

The "Space Metal" project demonstrates the framework's efficacy through a seamless merge of a physical forest location and a high-fidelity CGI vessel.

- **Physical Data Acquisition:** The lens-based lead captured 32-bit HDRIs and reference plates to provide accurate environmental lighting data.
- **Interactive On-Set Lighting:** The team utilized red practical strobes balanced with daylight on-set to create real-world "light wrap" on talent, ensuring the environment reacted to the virtual ship's light source before rendering.
- **Technical Synchronization:** The CGI artist utilized physical metadata (24mm lens at f/8, camera height 60 inches, distance to red lighting, 22 feet) to match optical distortion and depth-of-field within a unified ACES color pipeline.

4 OBSERVATIONS AND EDUCATIONAL GROWTH.

The results show "exponential growth" in student output. 3D artists report a significant decrease in the "plasticity" of their renders because they are now mimicking physical behaviors they've seen in person. Simultaneously, photography students gain an "advantage" in the job market by learning how to prep assets for a professional VFX pipeline.

5 Conclusion: Replicable Success

We will conclude by sharing a roadmap for other institutions to implement similar cross-departmental collaborations, including how to structure critiques and what hardware is essential to bridge the gap between the lens and the computer in a team-based class structure.



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